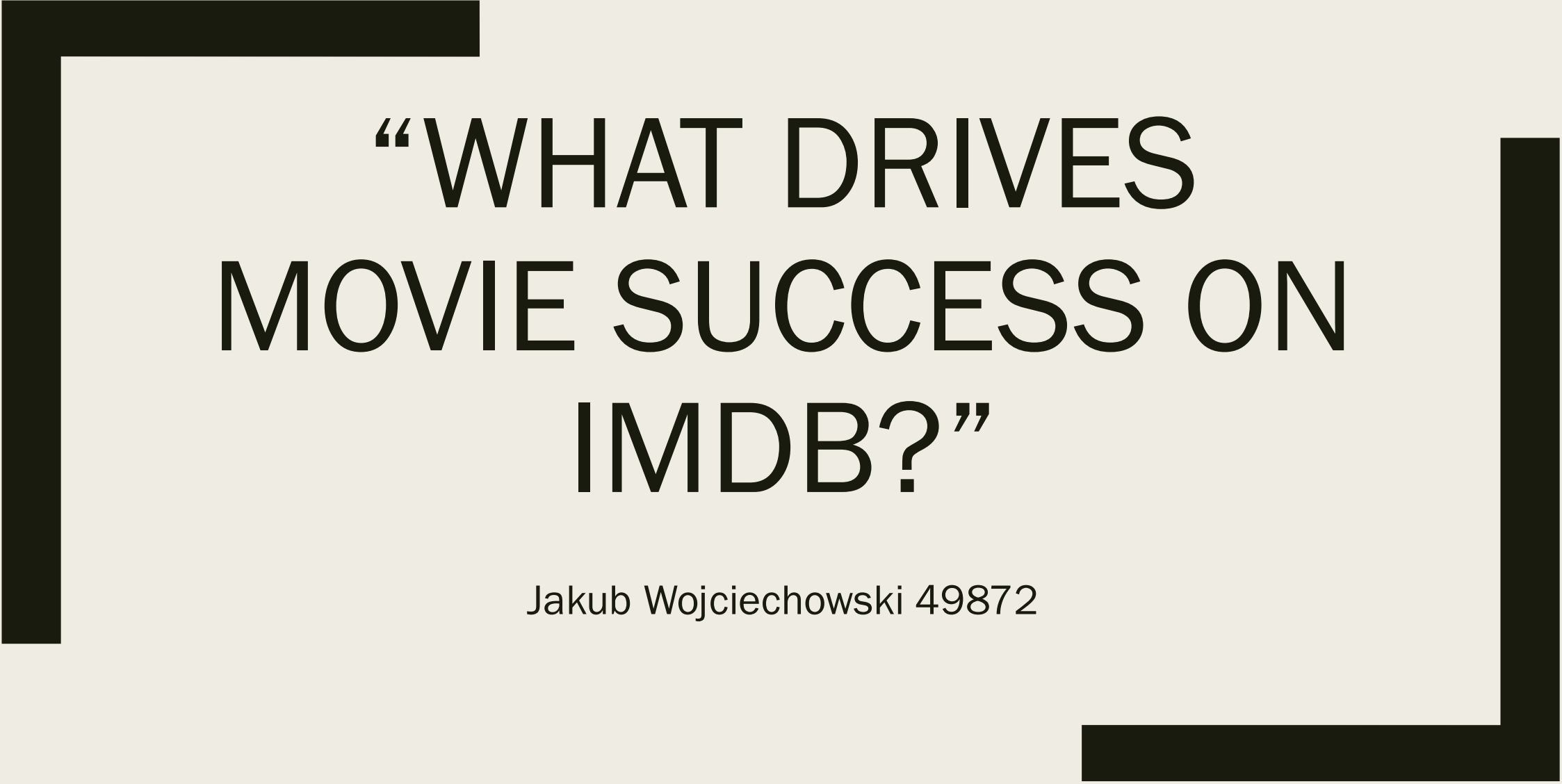


A thick black L-shaped frame is positioned on the left and bottom edges of the slide, framing the central text.

“WHAT DRIVES MOVIE SUCCESS ON IMDB?”

Jakub Wojciechowski 49872

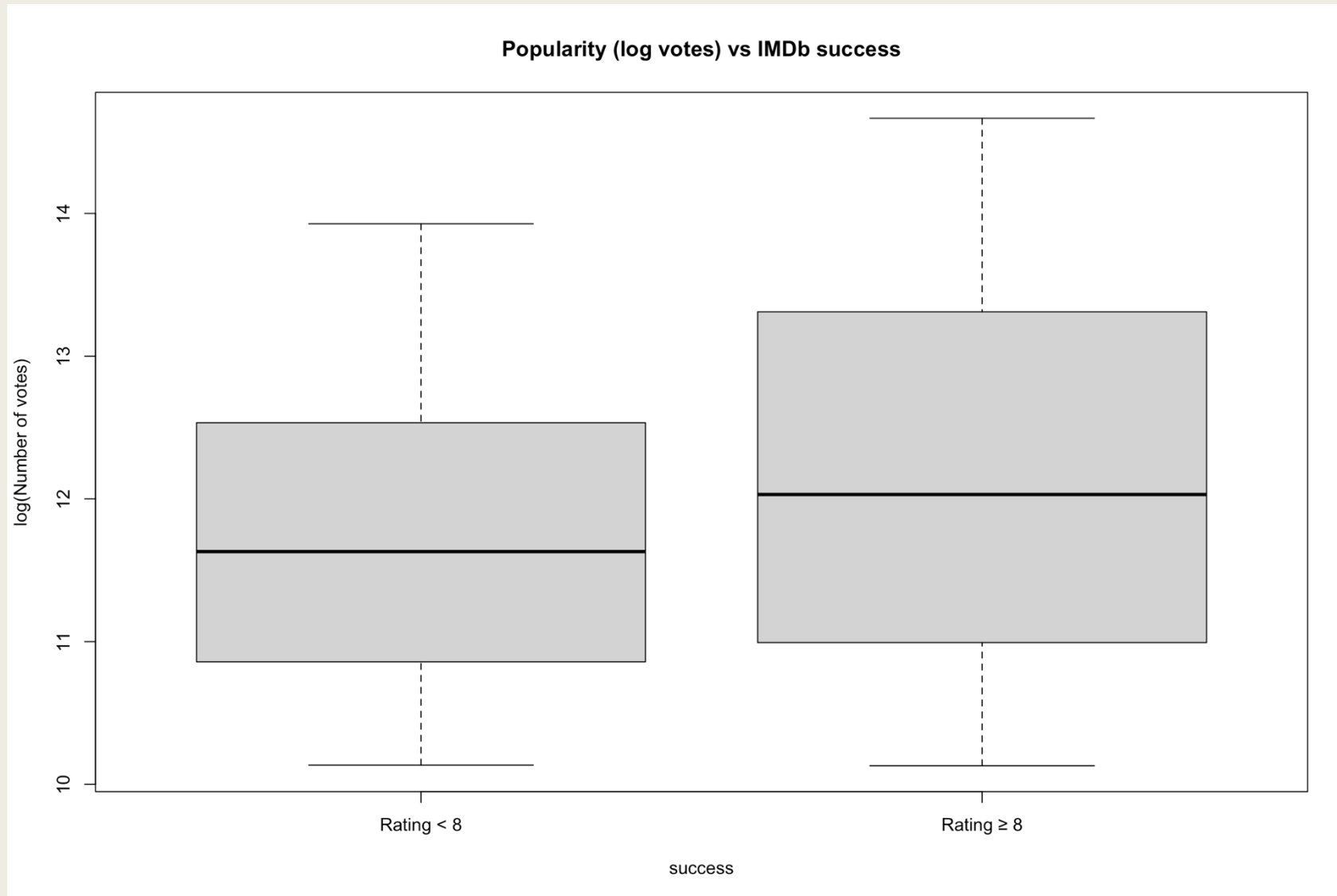
Basic information

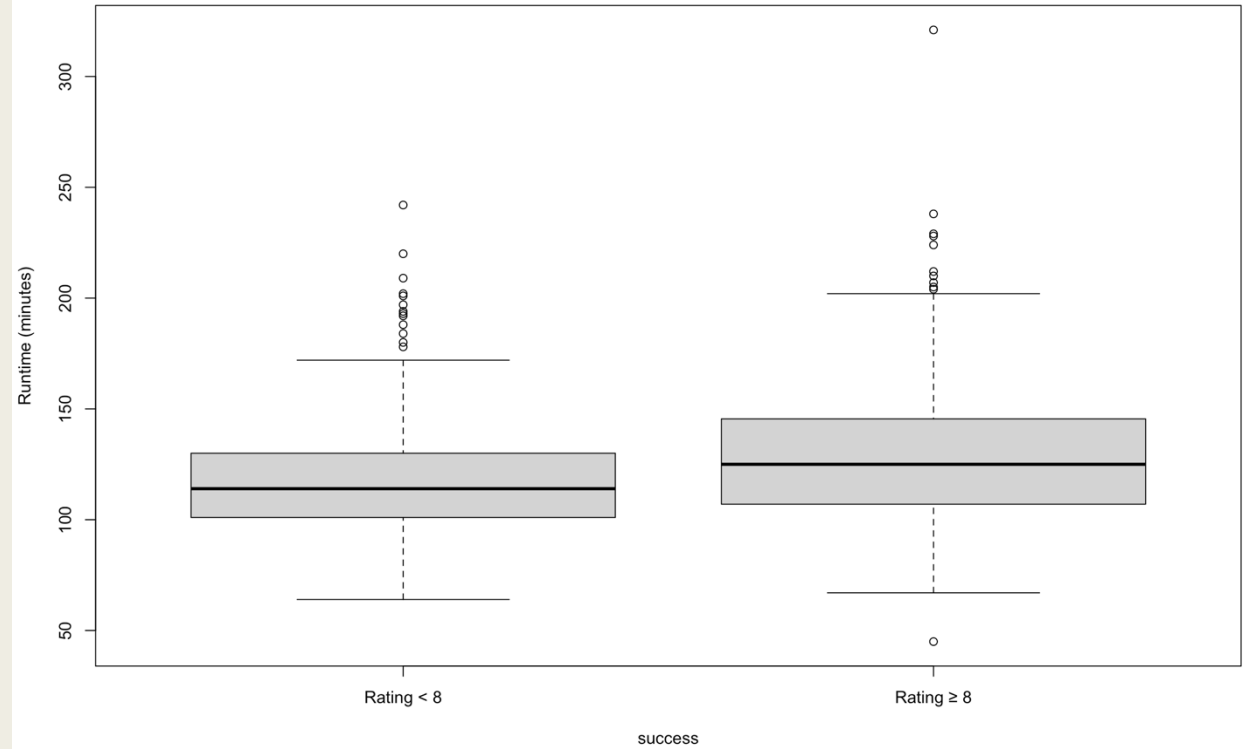
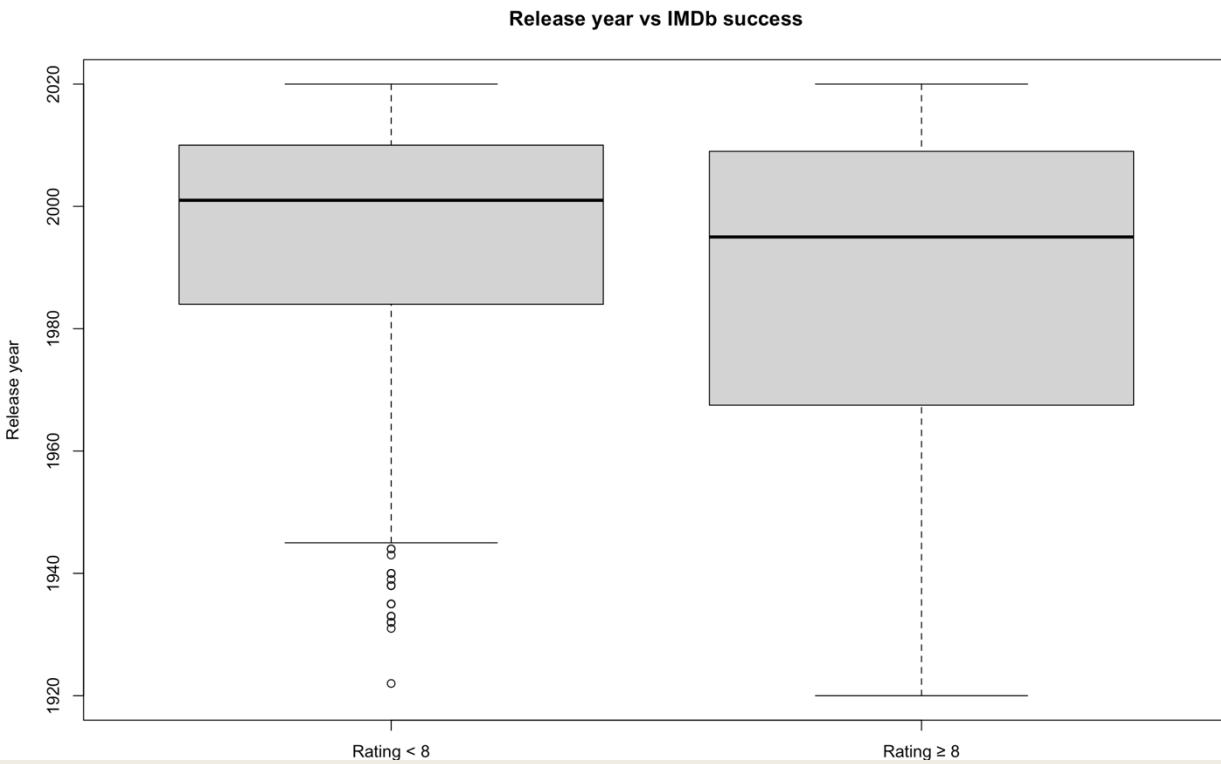
- Dataset: IMDb Top 1000 Movies
- Definition of success: IMDb rating ≥ 8.0

Data and variables

- 999 movies (after cleaning)
- Outcome variable: success (0/1)
- Key predictors:
 - popularity (number of votes, log-transformed)
 - runtime (minutes)
 - release year

Exploratory Data Analysis: Popularity





EDA: RUNTIME AND
RELEASE YEAR

Logistic regression model

- Logistic regression (binary outcome)
- Why logistic?
 - success is 0/1
- Predictors:
 - popularity
 - runtime
 - release year

Results and interpretation

- Popularity:

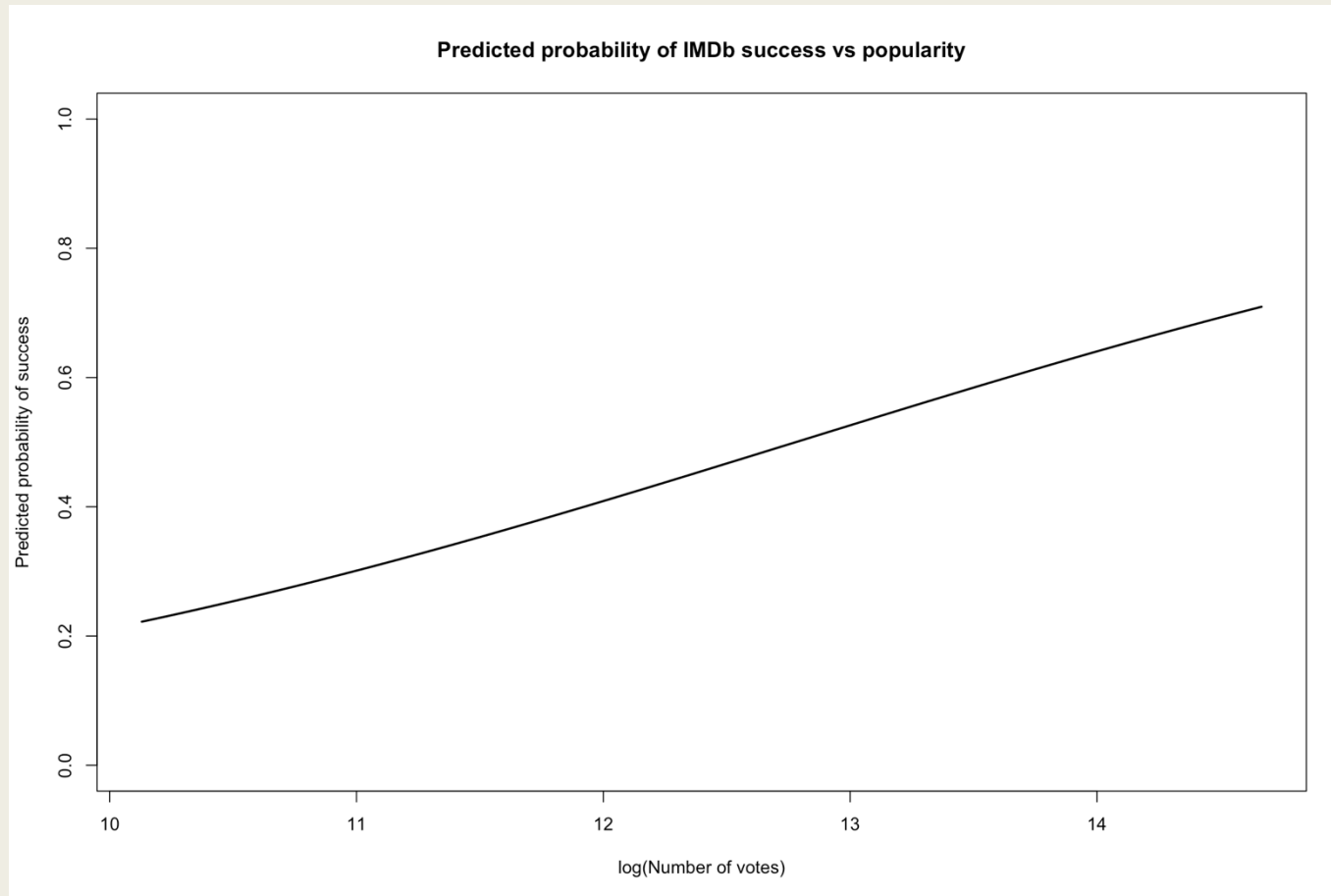
- $\sim 2.7\times$ more votes $\rightarrow$ $\sim 60\%$ higher odds of success

- Runtime:

- +10 minutes $\rightarrow$ $\sim 20\%$ higher odds

- Release year:

- newer movies $\rightarrow$ lower odds of success



Predicted probability of success

- 20k votes → ~22%
- 250k votes → ~45%
- 2M votes → ~70%

Conclusions and assumptions

■ Assumptions checked:

- correct model choice
- no multicollinearity ($VIF \approx 1$)

■ Final conclusions:

- popularity is key
- other effects are secondary

Thanks for your attention!